



Cyberster Blue Team Internship

WEEK 10 Report

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CSI-B1-487

Technical Objective

The objective of this phase was to perform a host-level digital forensic audit on a provided EWF bit-stream image ([Mantooth.E01](#)) to map compromise indicators, system structure, anti-forensic activity, and suspect intent. The scope of work involves evaluating structural image integrity, disk geometry, user configurations, hidden payloads, and file exhumation using automated forensic processing.

- **Analysis Workstation Path:** [Desktop/haris/autopsies](#)
- **Forensic Suite:** Autopsy Core Engine (v4.23.0)
- **Target Image Baseline:** [Mantooth.E01](#)

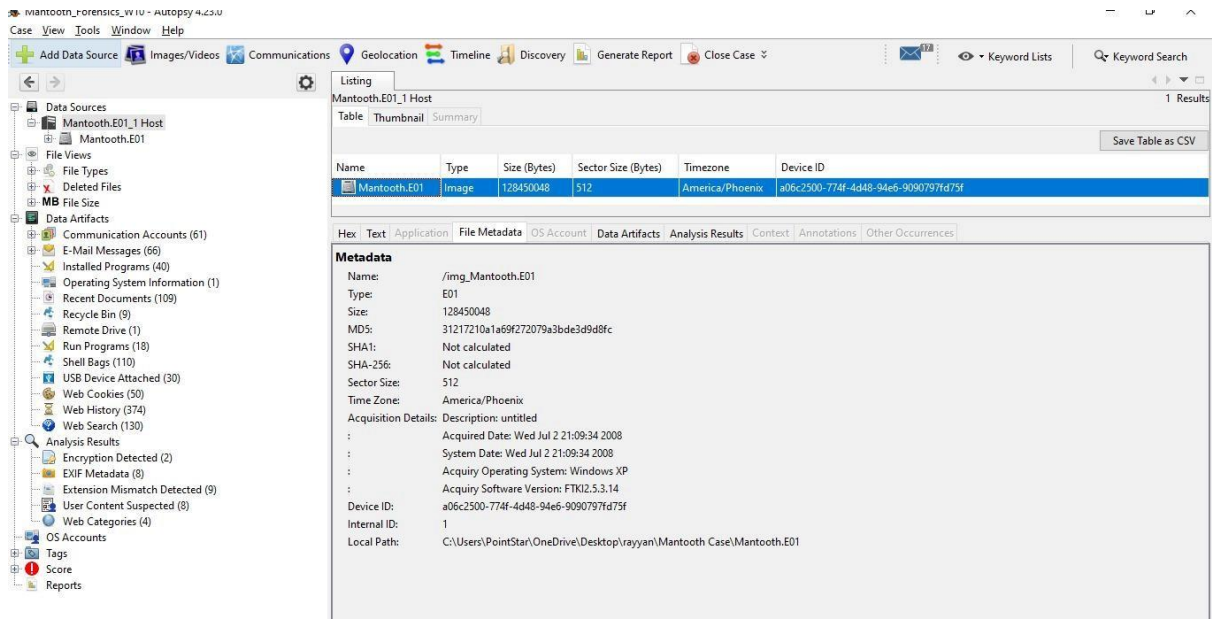
Part A: File System Triage & Evidence Extraction

In this phase of the investigation, we performed a structural validation and physical file layout assessment of the image data to isolate malicious staging areas and bypass anti-forensic concealment.

1. Data Ingestion & Image Integrity Verification

To ensure forensic preservation and adhere to strict chain of custody requirements, the bit-stream image was validated upon initial ingestion to guarantee zero data modification.

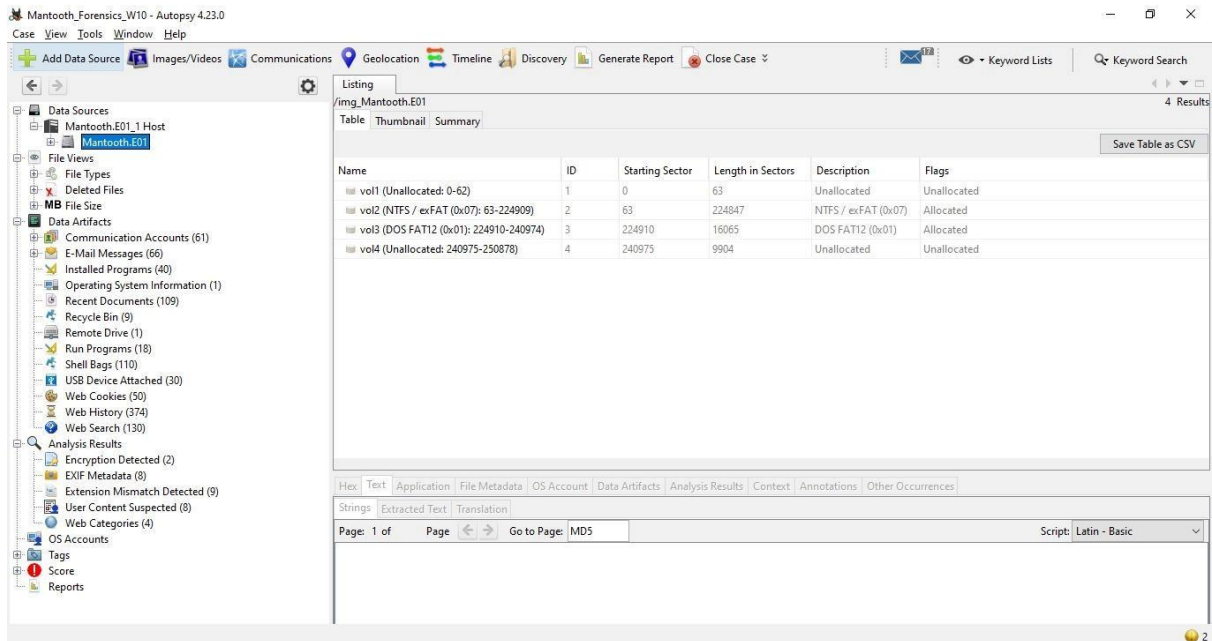
- **Action Performed:** Mounted the [Mantooth.E01](#) image file into the Autopsy case workspace and executed the Data Source Integrity ingest module.
- **Metrics & Findings:**
 - **Verified MD5 Hash:** [31217210a1a69f272079a3bde3d9d8fc](#)
 - **Source Artifact File:** [Reports/Data Source Integrity Report](#)
- **Status:** Integrity Confirmed. The internal database verification flags successfully matched the acquisition baseline perfectly, establishing that the digital evidence remained unaltered.



2. Disk Layout & Host Environment Triage

We analyzed the volume geometry of the target media to identify unallocated clusters and locate primary system directories.

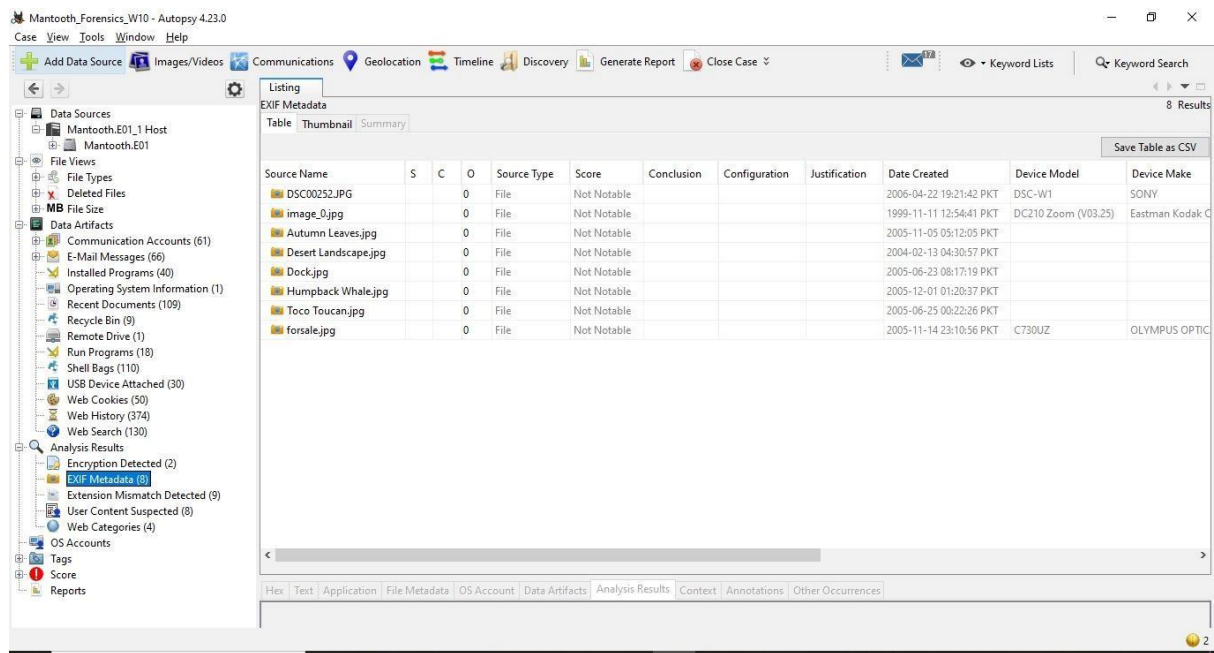
- **Action Performed:** Audited the storage partition mapping and queried the file system metadata layer.
- **Metrics & Findings:**
 - **Disk Partitioning:** Detected 4 distinct volumes (**vol1** through **vol4**).
 - **Primary OS Partition:** Localized directly on **vol2**, utilizing an NTFS file system starting at sector boundary Sector 63.
 - **Host Metadata:**
 - **Host Name:** **WESMANTOOTH-PC**
 - **Operating System Product Name:** **Windows Vista (TM) Ultimate**
 - **Processor Architecture:** **x86 (32-bit architecture)**
 - **Registered Owner:** **Wes Mantooth**
 - **Registered Organization:** **Volturi Enterprises**



3. EXIF Metadata Analysis & Evidence Tampering

EXIF tags were audited within media folders to authenticate the origin devices of captured graphic content and track physical location baselines.

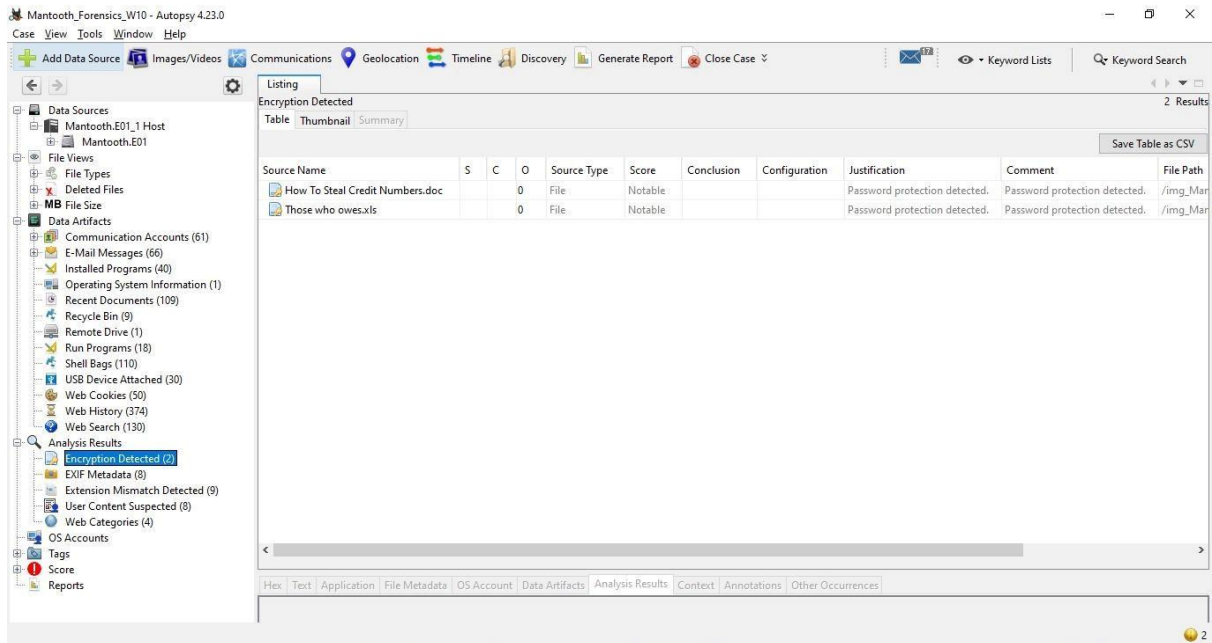
- **Action Performed:** Navigated to the **EXIF Metadata (8)** data artifact repository parsed from image files.
- **Metrics & Findings:**
 - **Target Artifact:** **forsale.jpg**
 - **Device Model:** **C730UZ**
 - **Device Make:** **OLYMPUS OPTICAL CO., LTD**
 - **Timestamp:** **2005-11-14 23:10:56 PKT**



4. Data Hiding & Encryption Detection

The file system was audited for cryptographic flags indicating password-protected containers or documents designed to conceal illicit text strings.

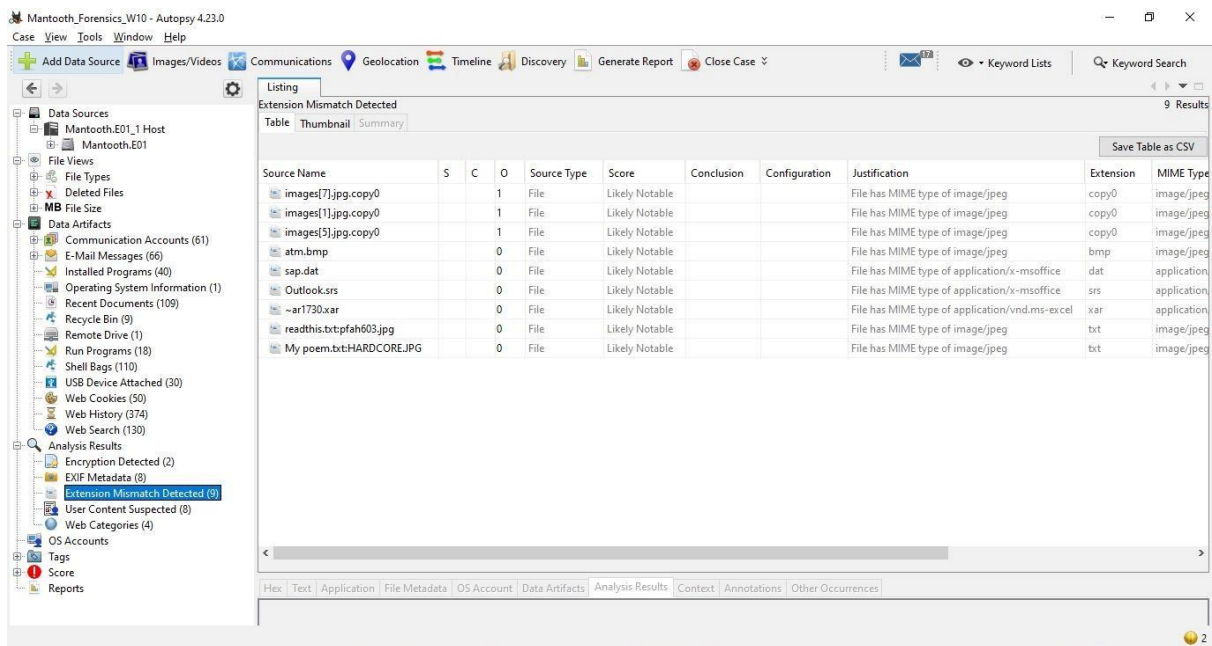
- **Action Performed:** Evaluated data streams flagged under the **Encryption Detected (2)** analysis view.
- **Metrics & Findings:** Autopsy automatically isolated 2 high-risk password-protected documents located inside the suspect's folders:
 - a. **How To Steal Credit Numbers.doc** (Password Protection Detected)
 - b. **Those who owes.xls** (Password Protection Detected)



5. Extension Mismatches & Alternate Data Streams (ADS)

We evaluated the consistency between raw file headers (MIME signatures) and user-facing extensions to intercept hidden files, alongside checking for hidden fork data structures.

- Action Performed:** Queried data artifacts generated inside the **Extension Mismatch Detected (9)** layout.
- Metrics & Findings:** Clear evidence of deliberate anti-forensic concealment was discovered:
 - sap.dat:** Masqueraded as an application data log file, but headers revealed a MIME type of **application/x-msoffice** (Microsoft Office file signature).
 - My poem.txt:HARDCORE.JPG:** Exploited an NTFS Alternate Data Stream (ADS). A complete hidden JPEG image payload (**image/jpeg**) was attached invisibly to a standard plain text file stub.
 - readthis.txt:pfah603.jpg:** Second instance of an illicit ADS configuration hiding an image file framework inside a text file container.



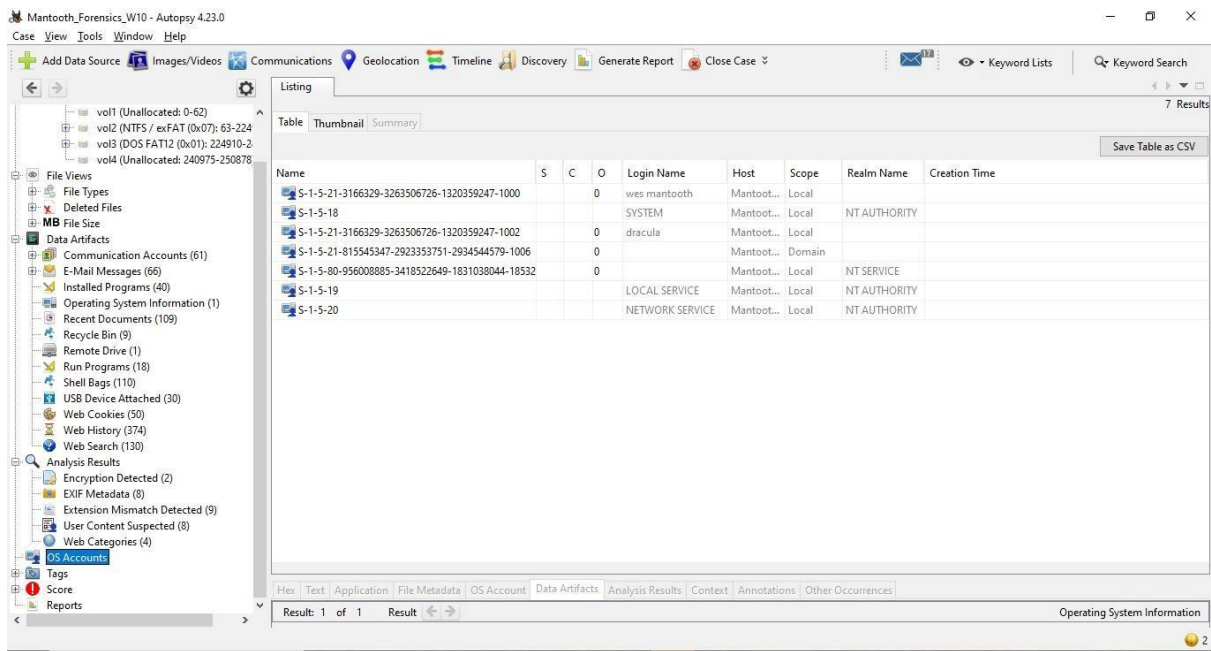
Part B: System Artifacts & Behavior Reconstruction

Part B shifts the focus from structural file layout to user actions, timeline reconstruction, account enumeration, and data exhumation.

1. User Account & SID Enumerate

Analysis of the **SAM** registry database was correlated to determine active local user profiles and evaluate privilege permissions.

- **Action Performed:** Extracted and indexed data artifacts under the **OS Accounts** schema branch.
- **Metrics & Findings:** Two specific active local security identifiers (SIDs) were tracked within the OS layer:
 - **Suspect Profile:** **wes mantooth** $\rightarrow$ **SID:** **S-1-5-21-3166329-3263506726-1320359247-1000**
 - **Co-Conspirator Profile:** **dracula** $\rightarrow$ **SID:** **S-1-5-21-3166329-3263506726-1320359247-1002**



2. File Deletion Tracking & Recycle Bin Exhumation

We audited unallocated database pointers and evaluated the recycle bin files to reconstruct the destruction timeline of system components.

- Action Performed:** Scanned the Deleted Files -> All directory tree and cross-referenced them directly with the parsed database inside the Recycle Bin (9) artifact view.
- Metrics & Findings:** The suspect tried to securely discard multiple highly suspicious utilities and files. Autopsy mapped the randomized \$D and \$R files back to their original names and paths:
 - CameraShy.exe** (Stored as \$R61QDFF.exe): Legitimate surveillance/hidden utility, deleted on 2007-07-14 22:55:57 PKT.
 - ValidateCreditCard.zip** (Stored as \$RNHBWN2.zip): Critical data staging component, deleted on 2007-07-14 22:57:06 PKT.
 - FileZilla_2_2_32_setup.exe** (Stored as \$RTHDU55.exe): FTP client utility used for data exfiltration, deleted on 2007-06-24 05:24:52 PKT.
 - Monkey Graphics Staging:** funny-monkey-3.jpg, monkey-nerd.jpg, and funny-monkey-2.jpg were dumped in a single batch on 2007-07-27 04:27:20 PKT.

Mantooth_Forensics_W10 - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing 276 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)
WAOLUSGM				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
WAOLUSGM				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
TOD				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
TOD				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
WER				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
WER				2007-07-14 03:13:52 PKT	2007-07-14 04:07:54 PKT	2007-07-14 03:13:52 PKT	2007-07-14 03:13:52 PKT	0	Unallocated
Support				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
Support			1	2007-07-14 04:00:10 PKT	2007-07-14 04:07:55 PKT	2007-07-14 04:00:10 PKT	2007-07-14 04:00:10 PKT	3414	Unallocated
WPD				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
WPD				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
Windows Live				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
SharingMetadata				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
CABVPJH6				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
CAEHINJ1				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
CAPBUSTC				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
down[]				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated
1YCAE8PLTSCA0GRWEFCAG6F8LNC406WRSBCAX			1	2007-07-13 04:17:25 PKT	2008-07-03 02:55:03 PKST	2008-07-03 02:55:00 PKST	2007-07-13 04:17:25 PKT	3816	Unallocated
7YCA2CRZGGCA3PUHVQCA6PWZ9MCA8XKOOHC			1	2007-07-13 04:17:25 PKT	2008-07-03 02:55:18 PKST	2008-07-03 02:55:16 PKST	2007-07-13 04:17:25 PKT	722	Unallocated
8ICAWOD3MWC1DEF0ZCA0ANW4TCA8SHDOKI			1	2007-07-13 04:17:26 PKT	2008-07-03 02:55:26 PKST	2008-07-03 02:55:24 PKST	2007-07-13 04:17:26 PKT	2205	Unallocated

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Mantooth_Forensics_W10 - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing 9 Results

Source Name	S	C	O	Path	Time Deleted	Username	Data Source
\$R2M7A26.jpg				C:\Users\Wes Mantooth\Pictures\funny-monkey-3.jpg	2007-07-27 04:27:20 PKT		Mantooth.E01
\$R61QDFF.exe				C:\Users\Wes Mantooth\Documents\CameraShy.exe	2007-07-14 22:55:57 PKT		Mantooth.E01
\$R9HZ0Z0.jpg				C:\Users\Wes Mantooth\Pictures\monkey-nerd.jpg	2007-07-27 04:27:20 PKT		Mantooth.E01
\$REZFRY8				C:\Users\Wes Mantooth\Pictures\Old Images	2007-07-26 04:41:23 PKT		Mantooth.E01
\$R49IB5.DLL				C:\Users\Wes Mantooth\Documents\Hacker Stuff\HE...	2007-07-14 22:56:05 PKT		Mantooth.E01
\$RQVPHB.jpg				C:\Users\Wes Mantooth\Pictures\funny-monkey-2.jpg	2007-07-27 04:27:20 PKT		Mantooth.E01
\$RKY3FVP.gif				C:\Users\Wes Mantooth\Pictures\Logo-ACE-100x100...	2007-07-27 04:27:20 PKT		Mantooth.E01
\$RNHBWN2.zip				C:\Users\Wes Mantooth\Documents\ValidateCreditCa...	2007-07-14 22:57:06 PKT		Mantooth.E01
\$RTHDU5.exe				C:\Users\Wes Mantooth\Desktop\FileZilla_2.2.32_setu...	2007-06-24 05:24:52 PKT		Mantooth.E01

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

3. Application Execution (Prefetch Analysis)

Prefetch binary structures (.pf) were evaluated to establish concrete proof of tool usage and calculate historical run counts.

- **Action Performed:** Analyzed application signatures inside the **Run Programs (18)** module.
- **Metrics & Findings:**

- **FTK IMAGER.EXE (FTK IMAGER.EXE-17AE1629.pf):** Executed a total of **38 times**, indicating heavy file manipulation/imaging workspace behavior, with a last run time of **2007-08-24 17:45:00 PKT**.
- **IEXPLORE.EXE (IEXPLORE.EXE-1B894AFB.pf):** Internet Explorer was executed **56 times**, confirming intensive browser activity, last running on **2007-08-24 18:06:48 PKT**.
- **CMD.EXE (CMD.EXE-89305D47.pf):** Manual command line terminal console was run **12 times** on the endpoint.

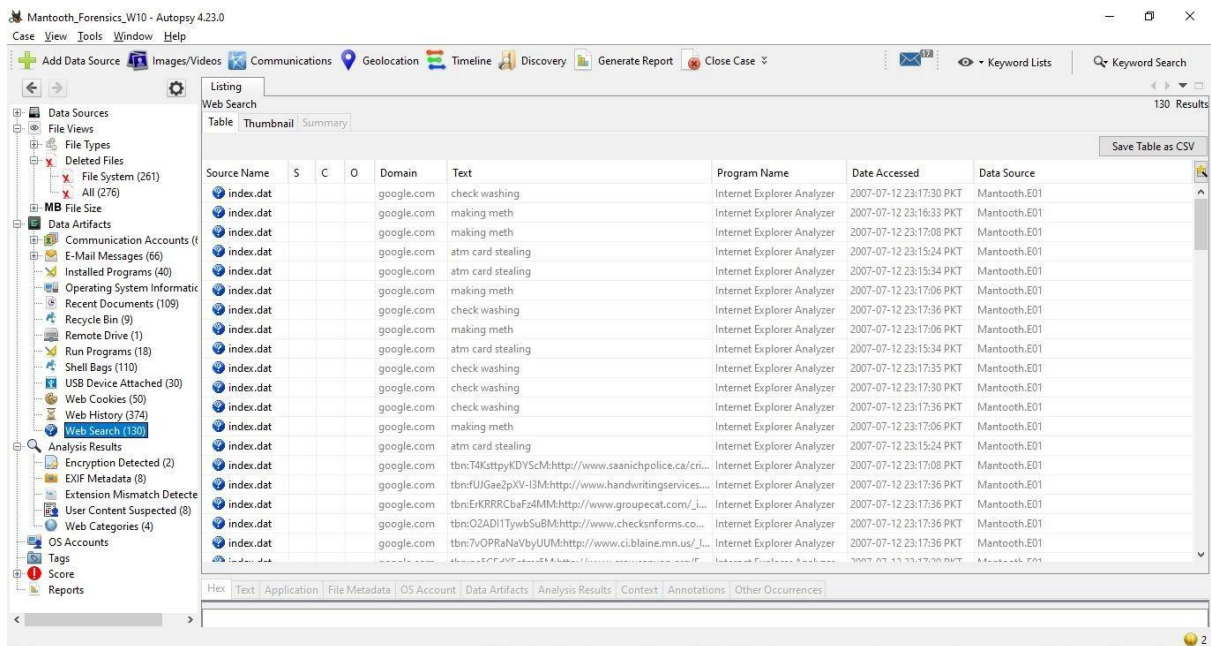
The screenshot shows the Autopsy 4.23.0 interface. The 'Run Programs' table is displayed, showing 18 results. The table columns are: Source Name, S, C, O, Program Name, Path, Date/Time, Count, and Comment. The results list various system executables and their execution times and counts.

Source Name	S	C	O	Program Name	Path	Date/Time	Count	Comment
AURORA.SCR-23204433.pf				AURORA.SCR	/WINDOWS/SYSTEM32	2007-08-24 20:10:46 PKT	18	Prefetch File
CMD.EXE-89305D47.pf				CMD.EXE	/WINDOWS/SYSTEM32	2007-08-24 17:38:24 PKT	12	Prefetch File
COMPMTGMLAUNCHER.EXE-0BF80059.pf				COMPMTGMLAUNCHER.EXE	/WINDOWS/SYSTEM32	2007-08-24 18:01:49 PKT	8	Prefetch File
CONSENT.EXE-69F6206D.pf				CONSENT.EXE	/WINDOWS/SYSTEM32	2007-09-27 18:10:25 PKT	140	Prefetch File
CONTROL.EXE-9459D5A0.pf				CONTROL.EXE	/WINDOWS/SYSTEM32	2007-08-24 15:54:24 PKT	18	Prefetch File
DEFRAG.EXE-738093E8.pf				DEFRAG.EXE	/WINDOWS/SYSTEM32	2007-09-27 17:09:36 PKT	35	Prefetch File
DFRGNTFS.EXE-4F838A89.pf				DFRGNTFS.EXE	/WINDOWS/SYSTEM32	2007-09-27 17:09:36 PKT	57	Prefetch File
DLLHOST.EXE-71214090.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32	2007-09-27 18:09:17 PKT	238	Prefetch File
DLLHOST.EXE-8930DF55.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32	2007-09-27 18:10:27 PKT	257	Prefetch File
DLLHOST.EXE-CB3D53F2.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32	2007-09-27 18:10:27 PKT	1	Prefetch File
DRVINST.EXE-5F8E77CD.pf				DRVINST.EXE	/WINDOWS/SYSTEM32	2007-08-24 15:09:27 PKT	18	Prefetch File
DWM.EXE-AEABE78B.pf				DWM.EXE	/WINDOWS/SYSTEM32	2007-09-27 16:16:51 PKT	1	Prefetch File
EFSSUI.EXE-DF03E0EF.pf				EFSSUI.EXE	/WINDOWS/SYSTEM32	2007-09-27 16:16:51 PKT	68	Prefetch File
EXPLORER.EXE-7A3328DA.pf				EXPLORER.EXE	/WINDOWS	2007-09-27 16:16:51 PKT	1	Prefetch File
FTK IMAGER.EXE-17AE1629.pf				FTK IMAGER.EXE	/PROGRAM FILES/ACCESSDATA/ACCESSDATA FTK IM...	2007-08-24 17:45:00 PKT	38	Prefetch File
IEUSER.EXE-D895AB54.pf				IEUSER.EXE	/PROGRAM FILES/INTERNET EXPLORER	2007-08-24 18:06:48 PKT	20	Prefetch File
IEXPLORE.EXE-1B894AFB.pf				IEXPLORE.EXE	/PROGRAM FILES/INTERNET EXPLORER	2007-08-24 18:06:48 PKT	56	Prefetch File
XCOPY.EXE-8E0707F2.pf				XCOPY.EXE	/WINDOWS/SYSTEM32	2007-08-24 17:47:33 PKT	15	Prefetch File

4. Intent Mapping via Browser Cache Searches

Browser keyword caches were audited to capture text entries directly indicating malicious intent or planning.

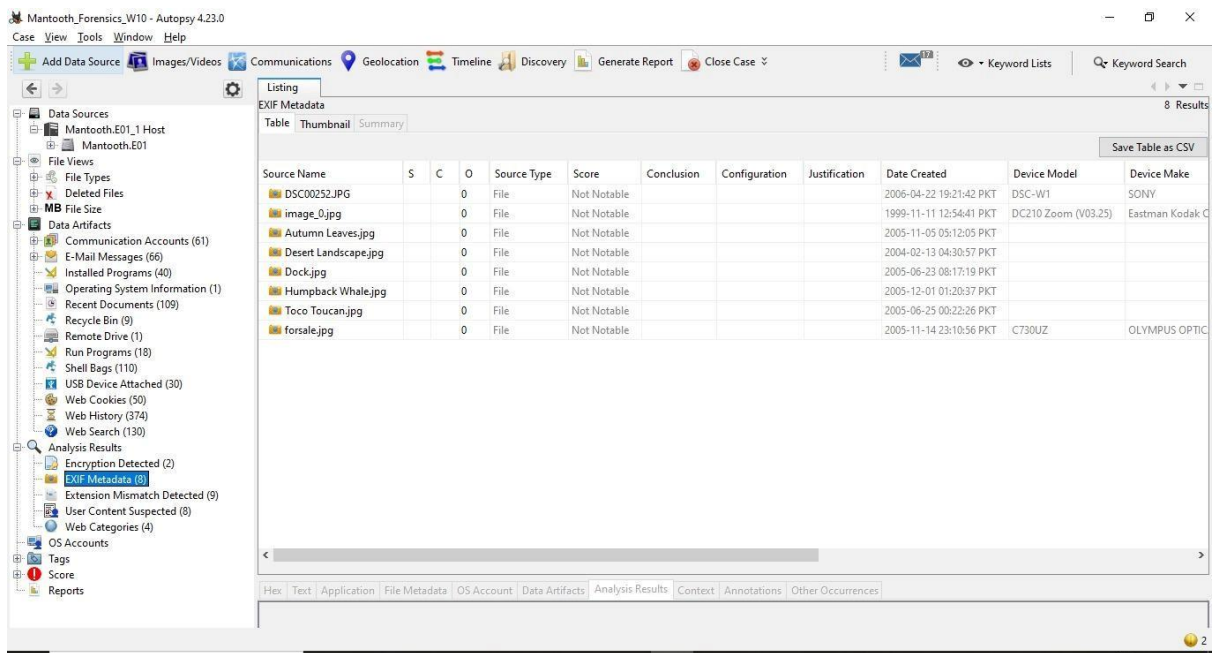
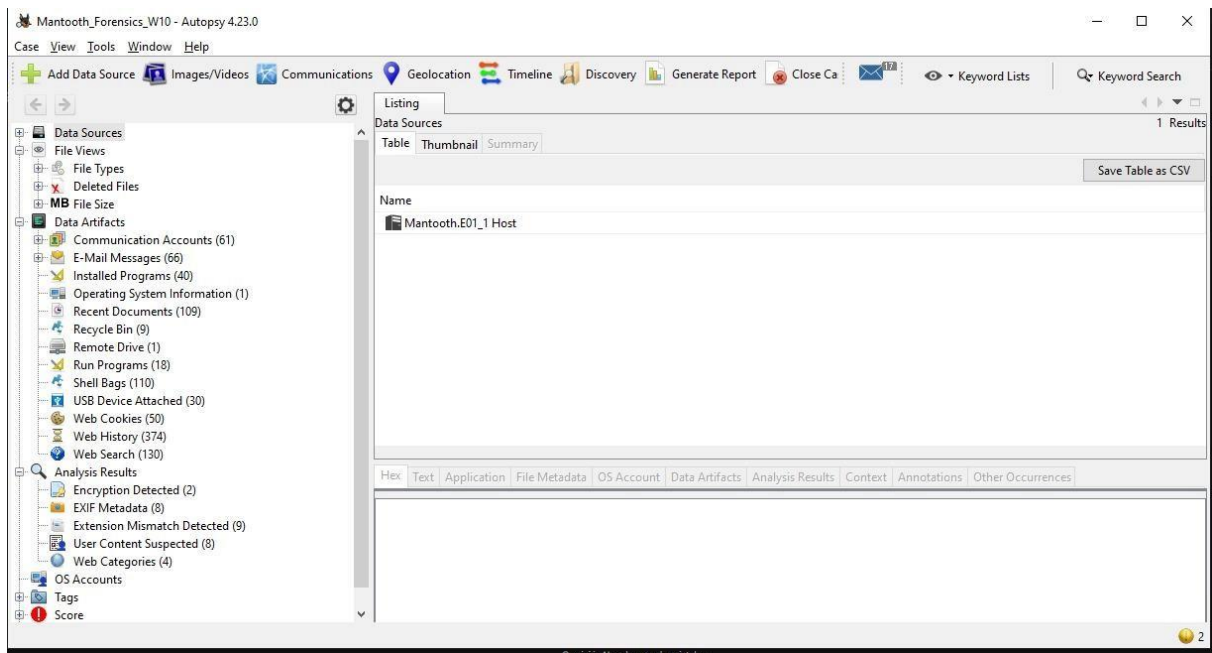
- **Action Performed:** Extracted strings from the **Web Search (130)** index.dat repository.
- **Metrics & Findings:** Google query engine tracking from **2007-07-12** between **23:15** and **23:17 PKT** verified clear criminal intent logs:
 - Financial Fraud: Searches for **atm card stealing**
 - Financial Forgery: Searches for **check washing**
 - Narcotics/Manufacturing: Repeated searches for **making meth**



5. Forensically Sound Registry Hive Staging

To conclude the triage, core configuration files were targeted for out-of-band extraction to preserve temporal registry states before zipping.

- Action Performed:** Manual data carving and extraction of root configuration files from `vol2/Windows/System32/config/` and `vol2/Users/mantooth/`.
- Extracted Artifact List:** `SYSTEM`, `SOFTWARE`, `SAM`, and `NTUSER.DAT`.
- Status:** Staged successfully to your folder workspace path for baseline archival and offline evaluation.



Final Comprehensive Forensic Timeline

The following timeline synthesizes all high-signal historical events captured across both Part A and Part B artifacts:

Date / Timestamp (PKT)	Artifact Source	Analytical Observation / Event Details
2005-11-14 23:10:56	EXIF Metadata / forsale.jpg	Image capture event on an Olympus C730UZ camera.
2007-06-24 05:24:52	Recycle Bin (\$RTHDU55.exe)	FTP client installer (FileZilla) manually thrown into the bin.
2007-07-12 23:15:34	Browser History / Web Search	Search sequence initiated for malicious card fraud concepts (atm card stealing).
2007-07-12 23:16:33	Browser History / Web Search	Search queries recorded for chemical compilation (making meth).
2007-07-12 23:17:30	Browser History / Web Search	Search logs captured for physical document alterations (check washing).
2007-07-14 22:55:57	Recycle Bin (\$R61QDFF.exe)	Deletion event for a critical monitoring application payload (CameraShy.exe).

2007-07-14 22:57:06	Recycle Bin (\$RNHBWN2.zip)	Financial data collection archive (ValidateCreditCard.zip) dropped in Recycle Bin.
2007-07-27 04:27:20	Recycle Bin (Multiple Images)	Batch deletion of hidden graphics resources (funny-monkey.jpg group items).
2007-08-24 17:38:24	Prefetch Tracking / CMD.EXE	Command prompt executed to verify file configurations or script actions.
2007-08-24 17:45:00	Prefetch Tracking / FTK IMAGER	Last tracked running session of FTK Imager on the host.
2007-08-24 18:06:48	Prefetch Tracking / IEXPLORE	Browser engine runtime ended right before target system seizure.